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Controllable differential system and vehicle having the controllable differential system

Abstract

A vehicle has a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame and having a lockable front differential; a manual override control; and a controller operatively connected to the lockable front differential and to the manual override control. The controller is adapted for controlling the motor to limit a speed of the motor when the front differential is locked and the manual override control is not activated.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5398565	12/1994	Brock	N/A	N/A
7600599	12/2008	Dahl et al.	N/A	N/A
8678976	12/2013	Wilson et al.	N/A	N/A
8790217	12/2013	Bowers et al.	N/A	N/A
9358882	12/2015	Hapka	N/A	N/A
9453570	12/2015	Martin, III	N/A	N/A
2009/0166106	12/2008	Batdorf	180/24.06	B60K 7/0007
2010/0262347	12/2009	Murota	701/69	B60K 23/08
2013/0167681	12/2012	Kll et al.	N/A	N/A
2019/0185080	12/2018	Bezeau-Tremblay et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102032336	12/2010	CN	N/A
104044567	12/2013	CN	N/A
105793094	12/2015	CN	N/A
109751383	12/2018	CN	N/A
102020000598	12/2019	DE	N/A
2012200154	12/2011	JP	N/A
WO-2013164656	12/2012	WO	B60K 23/04

OTHER PUBLICATIONS

International Search Report of PCT/IB2020/060215; Lee Young; May 24, 2021. cited by applicant
Office Action issued from the Chinese Patent Office on Apr. 4, 2025 in connection with the related
Application No. 202080091237.9. cited by applicant

Background/Summary

CROSS-REFERENCE (1) The present application is a divisional of U.S. patent application Ser. No. 17/772,764, filed Apr. 28, 2022, which is a 371 of International Patent Application No. PCT/IB2020/060215, filed Oct. 30, 2020, which claims priority from U.S. Provisional Application Ser. No. 62/928,793, filed on Oct. 31, 2019, the entirety of each of which is incorporated by reference herein.

TECHNOLOGICAL FIELD

(1) The present technology relates to a controllable differential system and to a vehicle having the controllable differential system.

BACKGROUND

(2) All-terrain vehicles (ATVs), side-by-side vehicles (SSVs) and similar vehicles are used for utility and recreational purposes. Such ATVs and SSVs are oftentimes provided with a transmission that can direct power from an internal combustion engine (ICE) to a rear drivetrain, for rear-wheel drive (or two-wheel drive) operation, or to both the rear drivetrain and a front drivetrain, for all-wheel drive (or four-wheel drive) operation.

(3) Some driving conditions, particularly when the vehicle is being driven off-road, may cause wheel slip, even when in all-wheel drive mode. Locking differentials may be mounted in either or both of the front and/or rear drivetrain. Conventional controls for locking and unlocking differentials are cumbersome. Some controls require stopping vehicles in order to lock or unlock their differentials.

(4) Thus, there is a desire for a controllable differential system and for a vehicle that addresses the aforementioned drawbacks.

SUMMARY

(5) It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.

(6) According to an aspect of the present technology, there is provided a controllable differential system for a vehicle, comprising: a differential having: a disengaged mode, and an engaged mode, the differential being selectively lockable while in the engaged mode; a differential lock switch having: a first position, a second position, and a third position, the second position being intermediate the first and third positions, the differential lock switch being biased to return to the second position when released from the third position, the differential lock switch being operatively connected to the differential for causing the differential to be: in the disengaged mode when the differential lock switch is in the first position, and in the engaged mode when the differential lock switch is in the second position; and a controller operatively connected to the differential and to the differential lock switch, the controller being adapted for controlling the differential to: selectively change the differential from one of an unlocked state and a locked state to another one of the unlocked state and the locked state when the differential lock switch is placed in the third position.

(7) In some implementations of the present technology, the controller is further adapted for controlling the differential to change from the one of the unlocked state and the locked state to the other one of the unlocked state and the locked state while the differential lock switch is being held in the third position for more than a predetermined time.

(8) In some implementations of the present technology, the controller is further adapted for controlling the differential to change from the one of the unlocked state and the locked state to the

other one of the unlocked state and the locked state when the differential lock switch is released from the third position after being held for more than a predetermined time.

(9) In some implementations of the present technology, the controllable differential system further comprises an actuator electrically connected to the differential lock switch and adapted to receive therefrom commands to engage and disengage the differential, the actuator being further electrically connected to the controller and adapted for receiving therefrom commands to lock and unlock the differential; and a mechanical lock adapted for being driven by the actuator to selectively engage and disengage the differential and to selectively lock and unlock the differential.

(10) In some implementations of the present technology, the actuator comprises an electric motor driven by an integrated circuit; and the mechanical lock comprises a rod driven by the electric motor and a lever driven by the rod, the lever being adapted to engage a sleeve in the differential to selectively disengage, engage or engage and lock the differential.

(11) In some implementations of the present technology, the actuator is adapted for providing an electronic feedback signal of the locked or unlocked state to the controller.

(12) In some implementations of the present technology, the differential is a front differential adapted for being mounted in a front drivetrain of the vehicle.

(13) According to another aspect of the present technology, there is provided a vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame, the front drivetrain comprising: a controllable differential system, comprising: a differential having: a disengaged mode, and an engaged mode, the differential being selectively lockable while in the engaged mode; a differential lock switch having: a first position, a second position, and a third position, the second position being intermediate the first and third positions, the differential lock switch being biased to return to the second position when released from the third position, the differential lock switch being operatively connected to the differential for causing the differential to be: in the disengaged mode when the differential lock switch is in the first position, and in the engaged mode when the differential lock switch is in the second position; and a controller operatively connected to the differential and to the differential lock switch, the controller being adapted for controlling the differential to: selectively change the differential from one of an unlocked state and a locked state to another one of the unlocked state and the locked state when the differential lock switch is placed in the third position; and a transmission adapted for directing power from the motor to the rear drivetrain and, when the differential is in the engaged mode, to direct power from the motor to the front drivetrain.

(14) According to a further aspect of the present technology, there is provided a vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame, the front drivetrain comprising a front differential selectably configurable in one of (i) a disengaged mode for rear-wheel drive operation of the vehicle, (ii) an engaged and unlocked mode, and (iii) an engaged and locked mode, the vehicle being operable in all-wheel drive when the front differential is engaged; a transmission adapted for: directing power from the motor to the rear drivetrain, and when the front differential is engaged, directing power from the motor to the front drivetrain; a controller operatively connected to the front differential; and a differential lock switch operatively connected to the front differential and to the controller, the differential lock switch being adapted for: providing a disengaging command to the front differential for placing the front differential in the disengaged mode, providing an engaging command to the front differential for placing the front differential in the engaged and unlocked mode, providing a locking command to the controller, the controller selectively causing the front differential to change from the engaged and unlocked mode to the engaged and locked mode when receiving the locking command, and providing an unlocking command to the controller, the controller causing the front differential to change from the engaged and locked mode to the engaged and unlocked mode when receiving the unlocking command.

(15) In some implementations of the present technology, the vehicle further comprises an actuator

operatively connected to the front differential and controlled by the controller, the actuator being adapted for: receiving the disengaging command from the differential lock switch; receiving the engaging command from the differential lock switch; receiving the locking command via the controller; receiving the unlocking command via the controller; and in response to receiving one of the disengaging command, the engaging command, the locking command and the unlocking command, selectively placing the front differential in a corresponding one of the disengaged mode, the engaged and unlocked mode and the engaged and locked mode.

(16) In some implementations of the present technology, the differential lock switch has: a first position, a second position, and a third position, the second position being intermediate the first and third positions, the differential lock switch being biased to return to the second position when released from the third position; and the differential lock switch is adapted for: providing the disengaging command when in the first position, providing the engaging command when moving from the first position to the second position, providing the locking command when placed in the third position while the front differential is in the engaged and unlocked mode, and providing the unlocking command when placed in the third position while the front differential is in the engaged and locked mode.

(17) In some implementations of the present technology, the locking command and the locking command are provided by the differential lock switch when the differential lock switch is held in the third position for more than a predetermined time.

(18) In some implementations of the present technology, the vehicle further comprises a brake control unit implementing an anti-lock braking system (ABS); the controller being further adapted for causing the brake control unit to disable the ABS when the front differential is in the engaged and locked mode.

(19) In some implementations of the present technology, the vehicle is an off-road vehicle selected from an all-terrain vehicle and a side-by-side vehicle.

(20) In some implementations of the present technology, the vehicle further comprises a handlebar pivotably connected to the frame and operatively connected to the front drivetrain, the differential lock switch being mounted on the handlebar.

(21) In some implementations of the present technology, the vehicle further comprises a manual override control; the controller being further adapted for controlling the motor to limit a speed of the vehicle when the front differential is in the engaged and locked mode and the manual override control is not activated.

(22) In some implementations of the present technology, the vehicle further comprises a manual override control; the controller being further adapted for controlling the motor to limit a speed of the motor when the front differential is in the engaged and locked mode and the manual override control is not activated.

(23) In some implementations of the present technology, the vehicle further comprises a handlebar pivotably connected to the frame and operatively connected to the front drivetrain, the manual override control being mounted on the handlebar.

(24) In some implementations of the present technology, the vehicle further comprises a steering assembly operatively connected to the front drivetrain; and a steering angle sensor operatively connected to the steering assembly; the controller being further operatively connected to the steering angle sensor and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when a steering angle indicated by the steering angle sensor exceeds a predetermined angle threshold.

(25) In some implementations of the present technology, the vehicle further comprises a vehicle speed sensor operatively connected to one of the front and rear drivetrains; the controller being further operatively connected to the vehicle speed sensor and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when a vehicle speed indicated by the vehicle speed sensor exceeds a predetermined

angle threshold.

(26) In some implementations of the present technology, the vehicle further comprises an engine speed sensor operatively connected to the motor; the controller being further operatively connected to the engine speed sensor and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when a speed of the motor indicated by the engine speed sensor exceeds a predetermined angle threshold.

(27) In some implementations of the present technology, the vehicle further comprises a gearbox sensor operatively connected to the transmission; the controller being further operatively connected to the gearbox sensor and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when the gearbox sensor indicates that the transmission is not in a predetermined state.

(28) In some implementations of the present technology, the vehicle further comprises a throttle operator operatively connected a throttle and to the motor; a throttle opening sensor; the controller being further operatively connected to the throttle opening sensor and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when an opening of the throttle reported by the throttle opening sensor exceeds a predetermined opening threshold.

(29) In some implementations of the present technology, the vehicle further comprises a controller area network (CAN) bus operatively connected to the controller; an indicator of a status of the CAN bus; the controller being further operatively connected to the CAN bus status indicator and being further adapted for delaying or preventing changing the front differential from the engaged and unlocked mode to the engaged and locked mode when the status of the CAN bus reported by the CAN bus status indicator indicates a loss of signal integrity.

(30) In some implementations of the present technology, the vehicle further comprises a visual display operatively connected to the controller, the controller being further adapted for: causing the visual display to display an indication when the front differential is prevented from changing from the engaged and unlocked mode to the engaged and locked mode; and causing the visual display to display an indication when changing from the engaged and unlocked mode to the engaged and locked mode is delayed.

(31) In some implementations of the present technology, the vehicle further comprises a visual display operatively connected to the controller and adapted for displaying a current disengaged mode, engaged and unlocked mode or engaged and locked mode of the front differential.

(32) In some implementations of the present technology, the visual display comprises a cluster of the vehicle.

(33) In some implementations of the present technology, the visual display comprises an icon appearing next to the differential lock switch.

(34) According to yet another aspect of the present technology, there is provided a vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame, the front drivetrain comprising a lockable front differential; a manual override control; and a controller operatively connected to the front differential and to the manual override control, the controller being adapted for controlling the motor to limit a speed of the motor when the front differential is locked and the manual override control is not activated.

(35) According to a still further aspect of the present technology, there is provided a vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame, the front drivetrain comprising a lockable front differential; a manual override control; and a controller operatively connected to the lockable front differential and to the manual override control, the controller being adapted for controlling the motor to limit a speed of the vehicle when the front differential is locked and the manual override control is not activated.

(36) In some implementations of the present technology, the vehicle further comprises a handlebar

pivotably connected to the frame and operatively connected to the front drivetrain, the manual override control being mounted on the handlebar.

(37) According to yet another aspect of the present technology, there is provided a vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame; a front drivetrain connected to the frame, the front drivetrain comprising a lockable front differential; a brake control unit implementing an anti-lock braking system (ABS); and a controller operatively connected to the front differential and to the brake control unit, the controller being adapted for causing the brake control unit to disable the ABS when the front differential is locked.

(38) In some implementations of the present technology, the front differential is selectably configurable in one of (i) a disengaged mode, (ii) an engaged and unlocked mode, and (iii) an engaged and locked mode, the vehicle further comprising a transmission adapted for directing power from the motor to the rear drivetrain and for selectably directing power to the front drivetrain.

(39) In some implementations of the present technology, the vehicle further comprises a differential lock switch operatively connected to the front differential and to the controller, the differential lock switch being adapted for: providing a disengaging command to the front differential for placing the front differential in the disengaged mode; providing an engaging command to the front differential for placing the front differential in the engaged and unlocked mode; providing a locking command to the controller, the controller selectively causing the front differential to change from the engaged and unlocked mode to the engaged and locked mode when receiving the locking command; and providing an unlocking command to the controller, the controller causing the front differential to change from the engaged and locked mode to the engaged and unlocked mode when receiving the unlocking command.

(40) For purposes of the present application, terms related to spatial orientation when referring to a vehicle and components in relation to the vehicle, such as “forwardly”, “rearwardly”, “left”, “right”, “above” and “below”, are as they would be understood by a driver of the vehicle sitting thereon in an upright driving position, with the vehicle steered straight-ahead.

(41) Embodiments of the present technology each have at least one of the above-mentioned object and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.

(42) Additional and/or alternative features, aspects, and advantages of embodiments of the present technology will become apparent from the following description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

(2) FIG. 1 is a perspective view, taken from a rear left side, of an all-terrain vehicle;

(3) FIG. 2 is a left side elevation view of the frame, engine, continuously variable transmission, seat, steering assembly and front and rear suspension assemblies of the all-terrain vehicle of FIG. 1;

(4) FIG. 3 is a perspective view, taken from a rear left side, of a steering assembly, front and rear drivetrains and brake assemblies of the all-terrain vehicle of FIG. 1;

(5) FIG. 4 is a right side elevation view of the steering assembly, the front and rear drivetrains and brake assemblies of the all-terrain vehicle of FIG. 1;

(6) FIG. 5 is a top plan view of the steering assembly, the front and rear drivetrains and brake

assemblies of the all-terrain vehicle of FIG. 1;

(7) FIG. 6 is a perspective view of an engageable and lockable front differential of the vehicle of FIG. 1 according to an embodiment;

(8) FIG. 7a is a perspective view of the differential of FIG. 6 with a cover removed for showing a mechanical lock driven by an actuator;

(9) FIG. 7b schematically shows details of the actuator of FIG. 7a;

(10) FIGS. 8a, 8b and 8c are cross-sectional views of the differential of FIG. 6 in disengaged mode, engaged and unlocked mode, and engaged and locked mode, respectively;

(11) FIG. 9 is a rear perspective view of a right switch control housing for the vehicle of FIG. 1 and of wires connecting the right switch control housing to the differential of FIG. 6 and to a controller according to an embodiment;

(12) FIGS. 10a, 10b and 10c are detailed rear perspective views of the right switch control housing of FIG. 9 showing a differential lock switch in first, second and third positions, respectively;

(13) FIG. 11 is a rear elevation view of a left switch control housing for the vehicle of FIG. 1, according to an embodiment;

(14) FIG. 12 is a block diagram showing connections of the front differential and of electrical components of the vehicle of FIG. 1 according to an embodiment;

(15) FIG. 13 is a block diagram showing internal components of a controller of FIG. 12 according to an embodiment; and

(16) FIG. 14 is an enlarged top plan view of a braking control unit.

DETAILED DESCRIPTION

(17) The present technology will be described with reference to a four-wheeled straddle-seat all-terrain vehicle (ATV) 9. However, it is contemplated that aspects of the present technology could be used in other types of vehicles, such as side-by-side vehicles (SSVs), dune buggies, and other off-road vehicles.

(18) With reference to FIGS. 1 to 5, the ATV 9 has a front end 2 and a rear end 4 defined consistently with a forward travel direction of the ATV 9. The ATV 9 has a frame, for example a tubular frame 12. An internal combustion engine 16 for powering the ATV 9 is connected to the frame. It is contemplated that the ATV 9 maybe powered by other types of motors, being for example powered by an electric motor.

(19) The ATV 9 has a front drivetrain 6 and a rear drivetrain 8, which are both connected to the frame 12. The front drivetrain 6 has two front wheels 18 and the rear drivetrain 8 has two rear wheels 18, although a rear drivetrain having a single wheel 18 or having four wheels 18 is also contemplated. Each wheel 18 is provided with low-pressure balloon tires adapted for off-road conditions and for traversing rugged terrain.

(20) The wheels 18 are operatively connected to the engine 16 via a transmission that, in the shown example, comprises a continuously variable transmission (CVT) 17 receiving power from the engine 16 and a gearbox 19 operatively connected to the CVT 17. A shifter 34 located underneath the right-hand side of a handlebar 32 is used by the driver to place the gearbox 19 in one of a high, low, neutral or reverse transmission positions. When a front differential 76 of the front drivetrain 6 is disengaged, the CVT 17 and the gearbox 19 direct power from the engine 16 to the rear wheels 18 only, allowing the ATV 9 to be operated in rear-wheel drive (or 2-wheel drive) mode. The CVT 17 and the gearbox 19 direct power from the engine 16 to both the rear wheels 18 and the front wheels 18 when the front differential 76 is engaged, allowing the ATV 9 to be operated in all-wheel drive (or 4-wheel drive) mode.

(21) As illustrated on FIG. 1, the ATV 9 also includes fairings 60 including a front fascia 62 at the front end 2 of the ATV 9 and several side panels 64 extending over lateral sides of the ATV 9. A fender 66 is disposed over each wheel 18 to protect a driver and/or passenger from dirt, water and other debris being projected by the rotating wheels 18. The ATV 9 further includes a straddle-type seat 28 mounted to the frame 12 for accommodating a driver of the ATV 9. Footrests 50 are

provided on either of the seat **28** and are disposed vertically lower than the seat **28** to support the driver's feet.

(22) The front wheels **18** are suspended from the frame **12** by left and right front suspension assemblies **24** while the rear wheels **18** are suspended from the frame **12** by left and right rear suspension assemblies **26**.

(23) A steering assembly **30** operatively connected to the front wheels **18** is rotationally supported by the frame **12** to enable the driver to steer the ATV **9**. The steering assembly **30** includes the handlebar **32** having a handle grip **33** at each end. The handlebar **32** is pivotably connected to the frame **12** and operatively connected to the front wheels **18** via a steering column **74**. Operation of the handlebar **32** causes a power steering assembly including a power steering gearbox **31** assisted by an electric motor **29** to actuate steering linkages **70** that are operably connected to left and right front wheels **18**. It is contemplated that the power steering electric motor **29** could be omitted. Other vehicles may include a steering wheel in place of the handlebar **32**.

(24) The ATV **9** can be operated in rear-wheel drive mode or in all-wheel drive mode. Referring to FIGS. **3**, **4** and **5**, the front drivetrain **6** of the ATV **9** includes the front differential **76**, a pair of half shafts **78** connected to constant velocity (CV) joints **128** and **88**, and shafts extending between the CV joints **88** and hubs **86**, each of these shafts being supported by a corresponding kingpin **85**. On each of its left and right sides, the front differential **76** is connected to respective CV joint **128** connected to one end of a respective half shaft **78**. Another CV joint **88** is connected to an opposed end of each half shaft **78**. Each CV joint **88** and **128** is protected by a respective boot cover **102**. A front wheel **18** and a front disc **82** are mounted to each hub **86** and are supported by the corresponding kingpin **85**. On each side of the front differential **76**, the front wheel **18**, the front disc **82**, the CV joint **88**, the half shaft **78** and the CV joint **128** are operably connected to rotate together, at a common speed.

(25) Left and right front brake assemblies include the front discs **82** and further include front calipers **84** mounted on the kingpins **85**. The left and right front calipers **84** are respectively connected to hoses **95L** and **95R**. Each front caliper **84** includes a pair of brake pads **87** positioned on opposite sides of the respective front disc **82**. Actuating the front calipers **84** by application of a fluid pressure in the hoses **95L** and/or **95R** causes the brake pads **87** to apply pressure on the respective front discs **82**.

(26) The rear drivetrain **8** of the ATV **9** includes a rear spool gear **96** enclosed within a housing **97**, an input shaft **100** of the spool gear **96**, CV joints **130** and **132**, half shafts **98** and shafts extending between the CV joints **132** and hubs **104**. The input shaft **100** is operably connected to a rear driveshaft **101** by a universal joint **81**. The rear driveshaft **101** is connected by another universal joint **81** to the gearbox **19** to receive power from the engine **16**. On each of its left and right sides, the spool gear **96** is connected to a CV joint **130** connected to one end of a respective half shaft **98**. Another CV joint **132** is connected to an opposed end of each half shaft **98**. The CV joints **130** and **132** are protected by respective boot covers **102**. The hubs **104** are supported by lower ends of rear suspension assemblies **26**. A rear wheel **18** is mounted to each hub **104**.

(27) The spool gear **96** transmits a torque received from the engine **16** to the left and right rear wheels **18**. The spool gear **96** causes the CV joints **130**, the half shafts **98**, the CV joints **132**, the wheel shafts **104** and the rear wheels **18** located on both left and right sides of the ATV **9**, as well as a single rear disc **106**, to rotate together, at a common speed.

(28) A single rear brake assembly includes the single rear disc **106**, mounted to the right hub **104**, and a single rear caliper **108**. Although the single rear disc **106** as illustrated is located on the right-hand side of the ATV **9**, mounting the single rear disc **106** on the left-hand side of the ATV **9** is also contemplated. The single rear caliper **108** is supported by a right rear suspension assembly **26**. The single rear caliper **108** is connected to a hose **116**. The single rear caliper **108** includes a pair of rear brake pads (not shown) positioned on opposite sides of the single rear disc **106**. Actuating the single rear caliper **108** by application of a fluid pressure in the hose **116** causes the rear brake pads

to apply pressure on the single rear disc **106**. No rear brake assembly is present on the left-hand side of the ATV **9**, where the left rear wheel **18** is supported by the left rear suspension assembly **26**. Providing rear brake assemblies on both sides of the rear drivetrain **8** of the ATV **9** is also contemplated.

(29) A user of the ATV **9** may actuate either or both of two (2) user actuated braking input devices, for example a hand lever **90** and a foot lever **110**, to independently or jointly deliver a braking command for slowing down or stopping the ATV **9**. The hand lever **90** and a front master cylinder **92** are mounted on the handlebar **32**. The front master cylinder **92** can be directly filled with braking oil. Actuating the hand lever **90** causes the front master cylinder **92** to transmit the braking command to a braking control unit **124**. In the implementation as shown, the braking command from the hand lever **90** is in the form of a fluid pressure in a hose **94** connected between the front master cylinder **92** and the braking control unit **124**. In another implementation, actuation of the hand lever **90** may cause the application of a tension on a cable to deliver the braking command to the braking control unit **124**. When the hand lever **90** is actuated, a signal is sent for turning on the brake lights.

(30) The foot lever **110** and a rear master cylinder **112** are mounted on a bracket **114** supported by a lower portion of the frame **12**. The rear master cylinder **112** receives braking oil via a hose **118** from the reservoir **120**. Actuating the foot lever **110** causes the rear master cylinder **112** to exert a fluid pressure in a hose **126**, also resulting in the delivery of the braking command to the braking control unit **124**. As in the case of the hand lever **90**, implementations of the foot lever **110** applying the braking command to the braking control unit **124** by the application of a tension on a cable are also contemplated. The braking command is proportional (either linearly or non-linearly) to the pressure applied on the foot lever **110**. When the foot lever **110** is actuated, a signal is sent for turning on the brake lights.

(31) The braking control unit **124** may receive the braking command from the front master cylinder **92**, via the hose **94**, or from the rear master cylinder **112**, via the hose **126**, or from both master cylinders **92** and **112** and accordingly apply front and/or rear braking of the ATV **9**. As shown on FIG. **14**, the braking control unit **124** comprises an electronic controller **156** that commands a motor **157** of a hydraulic pump **158** to direct pressure in the hoses **95L**, **95R** and **116**. The braking control unit **124** may implement an anti-lock braking system (ABS) in which wheel speed sensors (not shown) report speeds of each of the wheels **18**, or speeds of each of the half-shafts **78** and **98**, to the electronic controller **156** of the braking control unit **124**. To prevent locking of the wheels **18**, the braking control unit **124** may temporarily release the pressure applied by one or both of the front calipers **84** and/or by the rear caliper **108**, on the corresponding front discs **82** and/or rear disc **106**, of a wheel **18** found to be rotating at a significantly lower speed than the other wheels **18**.

(32) The front differential **76** can be controlled to adopt one of a disengaged mode and an engaged mode. The ATV **9** operates in rear-wheel drive mode when the front differential **76** is disengaged and in all-wheel drive mode when the front differential **76** is engaged. Additionally, the front differential may be selectively placed in an unlocked state and in a locked state while in the engaged mode. When in engaged mode, whether locked or unlocked, the front differential **76** receives at its input shaft **140** a torque from the engine **16** via a front propeller shaft **80** that is in turn operatively connected to the gearbox **19**. In turn, the front differential **76** transmits the torque to the front wheels **18** via the CV joints **128**, the front half shafts **78**, and the CV joints **88**. When unlocked, the front differential **76** may send unequal torque to the two front wheels **18** so that the left front wheel **18** and the right front wheel **18** may rotate at different speeds. When locked, the front differential **76** sends equal torque to the two front wheels **18** that both rotate at a same speed.

(33) Turning now to FIGS. **6**, **7a**, **7b** and **8a** to **8c**, the front differential **76** is contained in a casing **136** having a cover **138**, the cover **138** being removed on FIG. **7a**. The front differential **76** has an input shaft **140** operatively connected to the front propeller shaft **80**. Left and right output shafts **142** of the front differential **75** are operatively connected to respective half shafts **78** via respective

CV joints **128**. An actuator **144** drives a mechanical lock **146** for selectively disengaging, engaging or engaging and locking the front differential **76**. The actuator **144** has a pair of electrical connectors **148** for receiving commands for disengaging, engaging or engaging and locking the front differential **76**. As shown on FIG. **7b**, the actuator **144** comprises an electric motor **147** driven by an integrated circuit **149** receiving the commands from the electrical connectors **148**. It is contemplated that the actuator **144** may be implemented as a solenoid.

(34) Within the front differential **76**, a ring gear **300** is operatively connected to the input shaft **140** via a gear (not shown) on the input shaft **140** and rotates therewith. A planetary gear housing **302** also rotates with the ring gear **300** and the input shaft **140**. When the front differential **76** is disengaged, a left planetary gear **302L** contained in the planetary gear housing **302** is disengaged from the left output shaft **142** and no torque is transmitted to the left front wheel **18**. A right planetary gear **302R** is connected to the right output shaft **142**, but the disconnection between the left planetary gear **302L** and the left output shaft **142** results in no torque being transferred the front differential **76** to the right output shaft **142**. The output shafts **142**, the front half shafts **78** and the front wheels **18** thus rotate independently from the input shaft **140**. When the front differential **76** is engaged, the torque from the input shaft **140** is transmitted via the left and right planetary gears **302L** and **302R** to the output shafts **142**. As long as the front differential **76** is engaged and not locked, the left and right planetary gears **302L** and **302R** may transfer the torque unevenly between the left and right front output shafts **142**, depending on riding conditions of the ATV **9**. When the front differential **76** is engaged and locked, relative motion between the left and right output shafts **142** is prevented or entirely eliminated.

(35) In more details, the mechanical lock **146** includes a rod **150** driving a lever **152**. As shown on FIG. **8a**, when the front differential **76** is disengaged, the lever **152** is held in a first angular position by the actuator **144** and the rod **150** to place a sleeve **154** away from the ring gear **300**. In that position, the left planetary gear **302L** is disengaged from an intermediate shaft **304** connected to the left output shaft **142**. No torque is transmitted to the intermediate shaft **304** and to the left output shaft **142**, thereby disengaging the front differential **76**. FIG. **8b** shows that the rod **150** is rotated in a second angular position by the actuator **144**, causing the lever **152** to move the sleeve **154** closer to the ring gear **300**. In that position, the sleeve **154** connects the intermediate shaft **304** to the left planetary gear **302L**. Depending on a torque on each front wheel **18** from the ground, the torque received on the input shaft **140** is transferred by the front differential **76** to either or both of the front wheels **18**. In that position, the front differential **76** is engaged without being locked. FIG. **8c** shows that the rod **150** is rotated further in a third angular position by the actuator **144**. In this third position, the sleeve **154** directly engages the ring gear **300** to lock the intermediate shaft **304** to the ring gear **300**, thus locking the front differential **76**.

(36) When the differential **76** is disengaged, no connection is made between the input shaft **140** and the output shafts **142** so no torque is transferred from the engine **16**, the CVT **17** and the gearbox **19** to the output shafts **142**; in this case, no torque is applied on the front wheels **18** and the ATV **9** is driven in two-wheel drive mode, more specifically in rear-wheel drive mode. When the differential **76** is engaged while in the unlocked state, a torque is received at the input shaft **140** and transmitted to the output shafts **142**; in this case, the ATV **9** is driven in all-wheel drive mode. In this mode, unequal torque may be transmitted by the front differential **76** to the left and right wheels **18**, which may rotate at unequal speeds depending on driving conditions. When the differential **76** is engaged while in the locked state, the torque is received at the input shaft **40** and an equal torque is transmitted to the output shafts **142**; in this case, the ATV **9** is driven in all-wheel drive mode without allowing any speed difference between the front wheels **18**.

(37) The commands to selectively disengage or engage the front differential **76** are received at the actuator **144** from a differential lock switch **160** of a right switch control housing **162** mounted on a right-hand side of the handlebar **32**, and result from an actuation of the differential lock switch **160** by the user of the ATV **9**. The commands to change the front differential **76** between the unlocked

state and the locked state are received at the actuator **144** from a controller, for example an engine control unit (ECU) **200**, which translates another actuation of the differential lock switch **160** by the user of the ATV **9**. Referring to FIG. **9**, the right switch control housing **162** supports the differential lock switch **160**, a throttle operator **164**, and a visual display comprising icons **166**, **168** and **170** that allow visualizing a current mode of operation of the front differential **76**. The throttle operator **164** is connected to a throttle (not shown) of the engine **16** via a cable **173** that extends within a wire sheath **172**. It is contemplated that the cable **173** connecting the throttle operator **164** to the throttle may be replaced by a drive-by-wire signal in which commands from the throttle operator **164** may be electronically transmitted to the throttle. The differential lock switch **160** is connected to a plug **175** via a wire **174** that extends from the right switch control housing **162**. The plug **175** is connected to another plug **176**. Two wires **177** and **178** extend from the plug **176**. The wire **177** transmits control signals from the differential lock switch **160** directly to the actuator **144** (schematically illustrated on FIG. **9**) and the wire **178** transmits control signals from the differential lock switch **160** to the ECU **200** (also schematically illustrated on FIG. **9**). The wires **174**, **177** and **178** may implement a controller area network (CAN) bus. Other wires **179** connect the ECU **200** to other systems of the ATV **9**.

(38) The differential lock switch **160** may be placed in three distinct positions. First, second and third positions of the differential lock switch **160** are respectively shown on FIGS. **10a**, **10b** and **10c**. The second position is intermediate the first and third positions and the differential lock switch **160** is biased to return to the second position when released from the third position. As such, the driver may place the differential lock switch **160** in one of the first and second positions, following which the differential lock switch **160** maintains the position selected by the driver. If the driver depresses the differential lock switch **160** to place it in the third position and then releases the differential lock switch **160**, the differential lock switch **160** returns to the second position.

(39) When the differential lock switch **160** is in the first position, it sends a disengaging command to the actuator **144** to place the front differential **76** in disengaged mode. When the differential lock switch **160** is in the second position, it sends an engaging command to the actuator **144** to place the front differential **76** in engaged mode. As a result, moving the differential lock switch **160** from the first position to the second position places the front differential **76** in engaged mode without locking the front differential **76**. It may be noted that the disengaging and engaging commands are sent directly to the actuator **144** by the differential lock switch **160**, without passing through the ECU **200**.

(40) Moving the differential lock switch **160** from the second to the third position may selectively cause changing of the front differential **76** between the unlocked state and the locked state. Given that the third position may be reached from the second position, the front differential **76** is expected to already be in engaged mode when the differential lock switch **160** is placed in the third position. In the third position, the differential lock switch **160** may send a locking command or an unlocking command to the ECU **200**, requesting to change the front differential **76** from the unlocked state to the locked state or from the locked state to the unlocked state. To this end, a signal is sent from the differential lock switch **160** to the ECU **200** when the differential lock switch **160** is placed in the third position, the signal ending when the differential lock switch **160** is released from its third position and returns to the second position. The ECU **200** translates this signal into the locking command or to the unlocking command, depending on the current status of the front differential **76**. In an embodiment, the ECU **200** may immediately translate the signal received when the differential lock switch **160** is placed in the third position into the locking or unlocking command.

(41) Optionally, the ECU **200** may detect that the differential lock switch **160** is maintained in the third position for at least a predetermined time before translating this signal into the locking or unlocking command. In an embodiment, the ECU **200** sends a control signal to the actuator **144** to lock the front differential **76**, if previously in the unlocked state, or to unlock the front differential, if previously in the locked state. In this embodiment, the ECU **200** causes the actuator **144** to

change the unlocked or locked state of the front differential **76** while the differential lock switch **160** is still in the third position after the predetermined time, or as soon as the differential lock switch **160** is released and returns to the second position after having been held for at least the predetermined time.

(42) In another embodiment, changing the front differential **76** from the unlocked state to the locked state may be conditional. One or more additional conditions may cause the ECU **200** to delay acting upon the locking command or may entirely prevent the ECU **200** from acting upon the locking command. Some of these additional conditions are based on inputs provided to the ECU **200** by other components of the ATV **9**. For example, FIG. **11** shows a left switch control housing **180** mounted on a left-hand side of the handlebar **32**. The left switch control housing **180** supports a start button **182**, a stop switch **184**, an override button **186**, and a headlight switch **190**. The start button **182**, the stop switch **184** and the override button **186** are operatively connected to the ECU **200** and are operated by the driver of the ATV **9** for starting the ATV **9**, initiating ignition of the engine **16**, enforcing a maximum speed limit of the ATV **9** and disabling the maximum speed limit. The headlight switch **190** controls the headlights either directly or via the ECU **200**.

(43) FIG. **12** shows how the ECU **200** interacts with the actuator **144**, the differential lock switch **160** held by the right switch control housing **162**, the throttle operator **164**, and the icons **166**, **168**, **170**, and with other components of the ATV **9**, to control the front differential **76**. Some of the interactions between the ECU **200** and the other components shown on FIG. **12** may take place over a CAN bus **216**. The ECU **200** is connected to the differential lock switch **160**, the actuator **144**, the front differential **76**, the engine **16**, the brake control unit **124**, and the override button **186**, all of which have been introduced hereinbefore. The ECU **200** may also be connected to an engine speed sensor **202** operatively connected to the engine **16**, a vehicle speed sensor **204** operatively connected to at least one of front drivetrain **6** and the rear drivetrain **8**, a throttle opening sensor **206**, which is operatively connected to the throttle operator **164**, to the cable **173** or to the throttle valve (not shown) of the engine **16**, a gearbox sensor **208** operatively connected to the gearbox **19** or to the shifter **34**, a CAN bus status indicator **210** tracking signal integrity on the CAN bus **216**, and a steering angle sensor **212** operatively connected to the steering assembly **30**. In an embodiment, the throttle opening sensor **206** and/or the CAN bus status indicator **210** may be implemented in the form of software functions within the ECU **200**, given that information necessary allowing the ECU **200** to determine an opening of the throttle and to determine a status of the CAN bus **216** may already be available within the ECU **200**.

(44) The ECU **200** may further be connected to a visual display, such as cluster **214** mounted in view of the driver of the ATV **9**, the cluster **214** being adapted for displaying various information elements to the driver. Alternatively or in addition, the ECU **200** may be connected to the icons **166**, **168** and **170** of the right switch control housing **162**.

(45) The differential lock switch **160** sends commands **220** to the actuator **144** for engaging and disengaging the front differential **76**. The differential lock switch **160** also sends a signal **222** to the ECU **200** as a command to lock or unlock the differential **76** when placed in the third position. In one embodiment, the signal **222** is sent as soon as the differential lock switch **160** is placed in the third position and the ECU **200** immediately interprets the signal **222** as a command to engage or disengage the front differential **76**. In another embodiment, the signal **222** is sent continuously while the differential lock switch **160** is maintained in the third position and the ECU **200** interprets the signal **222** as the command to engage or disengage the front differential when the differential lock switch **160** has been held in the third position for at least the predetermined time, or when it is released after having been held in the third position for at least the predetermined time. In a further embodiment, the differential lock switch **160** only sends the signal **222** after it has been held in the third position for the predetermined time. The actuator **144** sends digital feedback information **224** to the cluster **214** to allow the cluster **214** to display a current status of the front differential **76**. When the ECU **200** determines that conditions are met for locking or unlocking the front

differential **76**, it sends a corresponding control signal **226** to the actuator **144**. In response, the actuator **144** may send an electronic feedback signal **228** to the ECU **200** to indicate whether the front differential **76** is in unlocked state or in locked state. Sending a mechanical feedback from the front differential **76** to the ECU **200** to indicate whether the front differential **76** is in unlocked or in locked state is also contemplated. The ECU **200** may compare the information contained in the electronic feedback signal **228** with a most recent control signal **226** to determine whether the most recent control signal **226** has effectively been properly acted upon by the actuator **144**. The ECU **200** may send information **230** related to the general operation of the ATV **9** or related to the operation of the front differential **76** to the cluster **214**. The driver may enter commands on the cluster **214**, these commands being relayed to the ECU **200** via signals **232**.

(46) When the ECU **200** detects the locking command provided by the differential lock switch **160**, it may check values provided by one or more of the engine speed sensor **202**, the vehicle speed sensor **204**, the throttle opening sensor **206**, the gearbox sensor **208**, the CAN bus status indicator **210**, and the steering angle sensor **212**. The ECU **200** may delay or prevent sending the control signal **226** requesting the actuator to lock the front differential **76** under some conditions. The control signal **226** may be delayed or prevented when the engine speed reported by the engine speed sensor **202** exceeds a predetermined engine speed threshold, for example 4000 RPM. The control signal **226** may also be delayed or prevented when the vehicle speed reported by the vehicle speed sensor **204** exceeds a predetermined vehicle speed threshold, for example 40 km/h.

(47) The control signal **226** may further be delayed or prevented when an opening of the throttle reported by the throttle opening sensor **206** exceeds a predetermined opening threshold, for example a 30% throttle opening. The control signal **226** may also be delayed or prevented when the gearbox sensor **208** indicates that the CVT **17** is not in a predetermined state, for example reverse gear or low forward gear. The ECU **200** may also delay or prevent sending the control signal **226** when the status of the CAN bus **216** reported by the CAN bus status indicator **210** indicates a loss of signal integrity on the CAN bus **216**. The ECU **200** may also delay or prevent sending the control signal **226** when a steering angle indicated by the steering angle sensor **212** exceeds a predetermined angle threshold, for example 25 degrees. The ECU **200** provides the control signal **226** requesting the actuator to lock the front differential **76** to the actuator **144** when none of these conditions is present. In an embodiment, any one of the above conditions may cause the ECU **200** to abandon the request to lock the front differential **76**. In another embodiment, the ECU **200** may delay the request to lock the front differential **76** and send to the actuator **144** the control signal **226** requesting the actuator to lock the front differential **76** when the conditions having prevented the locking are not longer present.

(48) Locking the front differential **76** may cause the ECU **200** to adapt other driving conditions of the ATV **9**. For example, the ECU **200** may cause the brake control unit **124** to disable the ABS when the front differential **76** is in the engaged and locked state. The ECU **200** may implement an engine speed limiting function and control the engine **16** to limit a speed of the engine **16** when the front differential **76** is in the engaged and locked state. The driver may increase an engine speed limit enforced by the engine speed limiting function by depressing the override button **186**. Successively depressing the override button **186** may lead to successive increases of the engine speed limit. Instead of the override button **186**, use of another equivalent manual override control is also contemplated. The ECU **200** may also implement a vehicle speed limiting function and control the engine **16** to limit a speed of the ATV **9** when the front differential **76** is in the engaged and locked state, unless the override button **186** has been depressed by the driver to disable the vehicle speed limiting function.

(49) The ECU **200** provides feedback signals **234** to the right switch control housing **162** about the current status of the front differential **76**, allowing the icons **166**, **168** and **170** shown on FIGS. **10a**, **10b** and **10c** to provide visual feedback to the driver about a current operating mode of the front differential **76**. The icon **166** is lit when the front differential **76** is disengaged, the ATV **9** being

operated in rear-wheel drive mode. The icon **168** is lit when the front differential **76** is engaged and not locked, the ATV **9** being operated in all-wheel drive mode. When the front differential **76** is engaged and locked, the icon **170** is lit. The icon **170** may flash on and off when the driver has requested locking the front differential **76** while other conditions of the ATV **9**, described hereinabove, prevent or delay locking of the front differential **76**. The ECU **200** controls lighting of the icons **166**, **168** and **170** according to the current status of the front differential **76**. Instead of or in addition to the visual display provided by the icons **166**, **168** and **170**, the ECU **200** may cause the cluster **214** to display one or more icons providing the same or equivalent information.

(50) As shown on FIG. **13**, the ECU **200** comprises a processor **240** (or a plurality of cooperating processors **240**) operatively connected to a memory device **242** (or a plurality of cooperating memory devices **242**), to an input device **244** and to an output device **246** allowing the ECU **200** to communicate with the other components of the ATV **9** shown on FIG. **12**. Although one input device **244** and one output device **246** are shown, the ECU **200** may comprise a plurality of input and output devices, or one or more input/output devices adapted for communicating with other components of the ATV **9**. The memory device **242** contains a non-transient memory **248** storing computer instructions that, when executed by the processor **240**, allow the ECU **200** to control locking and unlocking of the front differential **76**. The memory device **240** may also store a table **250** including various parameters, for example the predetermined time for the third position of the differential lock switch **160**, the predetermined opening threshold for the throttle, the predetermined state of the CVT **17**, and the predetermined steering angle threshold for the steering assembly **30**.

(51) The ATV **9** includes other components such as an air intake system, an exhaust system, radiators, a gas tank, a battery, a starter, and the like. As it is believed that these components would be readily recognized by one of ordinary skill in the art, further explanation and description of these components will not be provided herein.

(52) Modifications and improvements to the above-described embodiments of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame and operatively connected to the motor; a front drivetrain connected to the frame and operatively connected to the motor, the front drivetrain comprising a lockable front differential; a controller operatively connected to the lockable front differential; and a manual override control operatively connected to the controller, the controller being adapted for controlling the motor to limit a speed of the motor in response to the front differential being locked and the manual override control not being activated, the controller being adapted for disabling the limit to the speed of the motor in response to the manual override control being activated.
2. The vehicle of claim 1, further comprising a handlebar pivotably connected to the frame and operatively connected to the front drivetrain, the manual override control being mounted on the handlebar.
3. The vehicle of claim 1, wherein the front differential is selectively configurable in one of (i) a disengaged mode, (ii) an engaged and unlocked mode, and (iii) an engaged and locked mode; and the vehicle further comprises a transmission adapted for directing power from the motor to the rear drivetrain and for selectively directing power to the front drivetrain.
4. The vehicle of claim 3, further comprising a differential lock switch operatively connected to the front differential and to the controller, the differential lock switch being adapted for: providing a disengaging command to the front differential for placing the front differential in the disengaged

mode; providing an engaging command to the front differential for placing the front differential in the engaged and unlocked mode; providing a locking command to the controller, the controller selectively causing the front differential to change from the engaged and unlocked mode to the engaged and locked mode in response to receiving the locking command; and providing an unlocking command to the controller, the controller causing the front differential to change from the engaged and locked mode to the engaged and unlocked mode in response to receiving the unlocking command.

5. The vehicle of claim 1, wherein: the front differential has: a disengaged mode, and an engaged mode, the front differential is selectively lockable while in the engaged mode; and the vehicle further comprises a differential lock switch having: a first position, a second position, and a third position, the second position being intermediate the first and third positions, the differential lock switch being biased to return to the second position in response to being released from the third position, the differential lock switch being operatively connected to the front differential for causing the front differential to be: in the disengaged mode in response to the differential lock switch being in the first position, and in the engaged mode in response to the differential lock switch being in the second position; and the controller is operatively connected to the differential lock switch, the controller is adapted for controlling the front differential to: selectively change the front differential from one of an unlocked state and a locked state to another one of the unlocked state and the locked state in response to the differential lock switch being placed in the third position.

6. The vehicle of claim 5, wherein the controller is further adapted for controlling the front differential to change from the one of the unlocked state and the locked state to the other one of the unlocked state and the locked state while the differential lock switch is being held in the third position for more than a predetermined time.

7. The vehicle of claim 5, wherein the controller is further adapted for controlling the front differential to change from the one of the unlocked state and the locked state to the other one of the unlocked state and the locked state in response to the differential lock switch being released from the third position after being held for more than a predetermined time.

8. The vehicle of claim 5, further comprising: an actuator electrically connected to the differential lock switch and adapted to receive therefrom commands to engage and disengage the front differential, the actuator being further electrically connected to the controller and adapted for receiving therefrom commands to lock and unlock the front differential; and a mechanical lock being driven by the actuator to selectively engage and disengage the front differential and to selectively lock and unlock the front differential.

9. The vehicle of claim 8, wherein: the actuator comprises an electric motor driven by an integrated circuit; and the mechanical lock comprises a rod driven by the electric motor and a lever driven by the rod, the lever being adapted to engage a sleeve in the front differential to selectively disengage, engage or engage and lock the front differential.

10. The vehicle of claim 8, wherein the actuator is adapted for providing an electronic feedback signal of the locked or unlocked state to the controller.

11. The vehicle of claim 1, wherein: controlling the motor to limit the speed of the motor in response to the front differential being locked and the manual override control not being activated comprises: controlling the motor to limit the speed of the motor to a first speed of the motor in response to the front differential being locked and the manual override control not being activated; and the controller is further adapted for controlling the motor to permit the speed of the motor to reach a second speed of the motor in response to the front differential being locked and the manual override control being activated, the second speed of the motor being greater than the first speed of the motor.

12. A vehicle, comprising: a frame; a motor connected to the frame; a rear drivetrain connected to the frame and operatively connected to the motor; a front drivetrain connected to the frame and

operatively connected to the motor, the front drivetrain comprising a lockable front differential; a controller operatively connected to the lockable front differential; and a manual override control operatively connected to the controller, the controller being adapted for controlling the motor to limit a speed to a first speed in response to the front differential being locked and the manual override control not being activated, the speed being one of a motor speed and a vehicle speed, the controller being further adapted for controlling the motor to permit the speed to reach a second speed in response to the front differential being locked and the manual override control being activated, the second speed being greater than the first speed.

13. The vehicle of claim 12, further comprising a handlebar pivotably connected to the frame and operatively connected to the front drivetrain, the manual override control being mounted on the handlebar.

14. The vehicle of claim 12, wherein the front differential is selectively configurable in one of (i) a disengaged mode, (ii) an engaged and unlocked mode, and (iii) an engaged and locked mode; and the vehicle further comprises a transmission adapted for directing power from the motor to the rear drivetrain and for selectively directing power to the front drivetrain.

15. The vehicle of claim 14, further comprising a differential lock switch operatively connected to the front differential and to the controller, the differential lock switch being adapted for: providing a disengaging command to the front differential for placing the front differential in the disengaged mode; providing an engaging command to the front differential for placing the front differential in the engaged and unlocked mode; providing a locking command to the controller, the controller selectively causing the front differential to change from the engaged and unlocked mode to the engaged and locked mode in response to receiving the locking command; and providing an unlocking command to the controller, the controller causing the front differential to change from the engaged and locked mode to the engaged and unlocked mode in response to receiving the unlocking command.

16. The vehicle of claim 12, wherein: the front differential has: a disengaged mode, and an engaged mode, the front differential is selectively lockable while in the engaged mode; and the vehicle further comprises a differential lock switch having: a first position, a second position, and a third position, the second position being intermediate the first and third positions, the differential lock switch being biased to return to the second position in response to being released from the third position, the differential lock switch being operatively connected to the front differential for causing the front differential to be: in the disengaged mode in response to the differential lock switch being in the first position, and in the engaged mode in response to the differential lock switch being in the second position; and the controller is operatively connected to the differential lock switch, the controller is adapted for controlling the front differential to: selectively change the front differential from one of an unlocked state and a locked state to another one of the unlocked state and the locked state in response to the differential lock switch being placed in the third position.

17. The vehicle of claim 16, wherein the controller is further adapted for controlling the front differential to change from the one of the unlocked state and the locked state to the other one of the unlocked state and the locked state while the differential lock switch is being held in the third position for more than a predetermined time.

18. The vehicle of claim 16, wherein the controller is further adapted for controlling the front differential to change from the one of the unlocked state and the locked state to the other one of the unlocked state and the locked state in response to the differential lock switch being released from the third position after being held for more than a predetermined time.

19. The vehicle of claim 16, further comprising: an actuator electrically connected to the differential lock switch and adapted to receive therefrom commands to engage and disengage the front differential, the actuator being further electrically connected to the controller and adapted for receiving therefrom commands to lock and unlock the front differential; and a mechanical lock

being driven by the actuator to selectively engage and disengage the front differential and to selectively lock and unlock the front differential.

20. The vehicle of claim 19, wherein: the actuator comprises an electric motor driven by an integrated circuit; and the mechanical lock comprises a rod driven by the electric motor and a lever driven by the rod, the lever being adapted to engage a sleeve in the front differential to selectively disengage, engage or engage and lock the front differential.
